

Kai Barker

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EDUCATION

University of California, Santa Barbara

Statistics and Data Science B.S.

Santa Barbara, CA

Graduated June 2026

Major GPA: 3.90/4.0

Activities: Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club

Coursework: Linear Algebra, Optimization, Machine Learning, Stochastic Processes, Time Series, Adv. Statistical Models, Big Data

SKILLS, CERTIFICATES & INTERESTS

Languages & Tools: Python (pandas, NumPy, etc.), R (tidyverse, dplyr), Excel, SQL, HTML/CSS, Git, LaTeX, API Integration

Big Data & Engineering: Apache Spark, PySpark, GCP (BigQuery, Dataproc, GCS), ETL Pipelines, Parquet Storage, Distributed SQL

Machine Learning & Statistics: Scikit-Learn, PyTorch, PCA, XGBoost, Feature Selection (LASSO, NSC), Bootstrap/Permutation Testing

Visualization & Reporting: Tableau, Plotly, Seaborn, Matplotlib, ggplot2, Interactive Dashboards, Powerpoint

Certificates: JP Morgan Chase Software Engineering, JP Morgan Chase Quantitative Analysis, HackerRank SQL

Interests: Basketball, Hiking, Backpacking, Rock Climbing, Cooking, Running, Pickleball

EXPERIENCE

Research Assistant

Foundations of Financial Technology Lab

Santa Barbara, CA

September 2024 - Present

- Analyzed 20M+ Ethereum blocks and 100M+ wallets to quantify wealth distribution, calculated a Gini coefficient of 0.96.
- Produced visualizations tracking inequality evolution over 5 years, revealing the top 1% of wallets control 95%+ of Ether supply.
- Filtered institutional and Sybil addresses to isolate organic wallet behavior and improve the accuracy of distribution metrics.
- Acknowledged in a published paper on blockchain wealth inequality with researchers from Yale, Cornell, and UCSB.

Engagement Manager and Business Analyst

Consult Your Community

Santa Barbara, CA

October 2025 - June 2026

- Led a 16-week EdTech market entry engagement for an AI tutoring startup as Engagement Manager, directing a 7-person team across 3 workstreams to deliver a market entry strategy targeting the client's first paid B2B pilot.
- Built financial models and Tableau dashboards across two engagements, modeling \$26M in recoverable funding across 115 CA community colleges, and communicating findings directly to non-technical clients to drive recommendations.
- Extracted and merged transactional data from Shopify, Square, and supplementary client records, normalizing corporate accounts by KPIs (average order value, dormancy periods, account activity, lifetime value) to identify \$40,000 in recoverable annual revenue.
- Translated data findings into actionable client deliverables including a CRM system, automated email reactivation campaigns, and a buyer-ready SOP to support acquisition preparation.

Quantitative Researcher

Department of Statistics and Applied Probability

Santa Barbara, CA

September 2024 - June 2026

- Analyzed 80,000+ congressional trade disclosures to identify statistically significant abnormal returns, validating a weighted alpha signal for portfolio construction.
- Applied Random Matrix Theory (Marchenko-Pastur) and Ledoit-Wolf shrinkage to clean a 560-asset covariance matrix, isolating 30 signal eigenvalues explaining 60.1% of variance.
- Optimized mean-variance portfolios in Python using cvxpy, achieving a 13.81% annual backtest return and +1.07% mean alpha (excluding 2022) vs. SPY over an 8-year walk-forward period.

PROJECTS

Gene Therapy Off-Target Effect Analysis

- Built end-to-end ETL pipeline on GCP using PySpark, Dataproc, and BigQuery SQL to process a 78,893-gene expression matrix and 705,000+ disease association records across 22 candidate zinc finger gene therapy treatments for Angelman syndrome research.
- Engineered distributed statistical validation and off-target burden scoring framework, parallelizing 1,800+ model fits across bootstrap replicates; identified three significance tiers ($p < 0.001$) and the safest Pareto-optimal treatment candidate.

NeurIPS Ariel Data Challenge

- Built a signal processing pipeline (sigma-clipped pixel masking, dark/flat field correction) to calibrate 200+ GB of raw telescope data across 14,000 files and 1,100+ exoplanets.
- Engineered a weighted ensemble of Gaussian Process Regression, XGBoost, and Ridge to predict atmospheric spectra across 283 wavelength channels, calibrating prediction uncertainty via GLL-optimal grid search.

MCMC & Simulated Annealing - Optimization Research

- Implemented Metropolis-Hastings MCMC in Python for message decryption, constructing the target likelihood from letter-to-letter transition probabilities trained on War and Peace text.
- Extended the algorithm with a simulated annealing modification (exponential cooling schedule, $\alpha=0.995$) over 50,000 iterations, reducing average incorrect letters from 22.3 to 21.6 vs. baseline.